through the sweat pore. At the other end, in the deeper part of the dermis, the tube forms a coiled mass with a blind end. The sweat passes through the sweat pore and evaporates from the surface by taking heat from the skin. The sweat glands are present in large amounts on the palms, soles, forehead and in armpits.

The sebaceous glands are irregularly shaped sac like glands that open into the hair follicles. The oily secretions [sebum] of these glands make the hair, water, proof and protect the skin from drying effects of the atmosphere due to high temperatures and low humidity.

## 2.12 REPRODUCTIVE SYSTEM

"Without the reproductive system. The human species could not survive. However, this system unlike other organs systems is not necessary for the survival of individual humans", but to produce a new individual".

The reproductive system is a collection of internal and external organs in both male and female.

### SEX ORGANS

| | Male sex organs | Female sex organs |
|---|---|---|
| Primary | A pair of testes | A pair of ovaries |
| Secondary | Epididymis gland<br>Vas deferens<br>Seminal vesicles<br>Prostrate gland<br>Urethra<br>Penis | Fallopian tubes<br>Uterus<br>Vaginal canal<br>Vagina<br>The breasts |

The smallest cell in man's body are sperm cell. (size 0.05 mm)

Testes: There is one pair of testes lying one in each scrotal sac. It weighs about 15 gms.

Scrotum: The scrotum is a bag of skin having two compartments one for each testis. The semiferous tubules and interstitial cells are concerned with the process of spermatogenesis and secretion of testosterone (male sex hormone).

Epididymis: The epididymis is a long, coiled tube that rests on the backside of each testicle. It functions in the carrying and storage of the sperm cells that are produced in the testes.

Vas deferens: This is a fibro elastic (30-40 cms) which extends from epididymis to end in ejaculatory duct. It is joined by seminal duct and opens in prostatic urethra.

The seminal vesicles: These are little sacs one on each side of the urethra near the base of the bladder. They also add a fluid to the "semen" to the stored sperm.

The prostate gland: The gland lies at the base of the urinary bladder, this adds another fluid to the semen which makes the sperms active in swimming to reach the ovum.

The penis: This is an external organ for both urinary and reproductive system.

Urethral openings: This is the tube that takes the sperm outside the body during ejaculation.

Male sex hormones: Androgens [maintains spermatogenesis and sexual activity].

Testosterone [stimulates secondary sexual characteristics].



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